Fundamentals of Geriatric Medicine

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A Case-Based Approach

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For our students: past, present, future . . .

Foreword

Prior to the evolution of modern medicine, with its superabundance of diagnostic and therapeutic medical technology and the rise of the litigious society, the value of clinical skills was evident in both history taking and the physical examination. Even today, physicians can make a correct diagnosis solely by utilizing their clinical skills in about 90% of patient encounters. Furthermore, in the past physicians understood their role as a "psychologist" and were more apt to be familiar with the social context of their patients. House calls were common. The doctor was also a "placebo" who, at his best, inspired hope and probably sped recovery.

Geriatricians use both clinical skills and take advantage of modern technology sparingly, for they know they are dealing with the most challenging and frail of patients—older patients who so often present with multiple, complex, interacting behavioral, social, and physical problems. In contrast to medicine for young people, working with the older patient is much more demanding. Furthermore, the complex issue of societal attitudes toward old people can come into play, specifically the physician's need to deal with natural fears of aging, dependency, depression, dementia, and death. Ageism is the enemy of effective medical treatment.

At its best, geriatrics exemplifies ways that medical care for all ages can become more humane, problem-oriented, and holistic.

Fundamentals of Geriatric Medicine: A Case-Based Approach discusses 32 core topics of geriatric medicine. Its case-based approach focuses ultimately upon the challenge of decision making. Soriano et al. have given us all a distinctive book that will further catalyze the emerging field of geriatric medicine.

Robert N. Butler, MD
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Preface

Fundamentals of Geriatric Medicine: A Case-Based Approach is intended as a practical educational companion to *Geriatric Medicine*, 4th edition, by Christine K. Cassel et al. (New York: Springer, 2003). This book covers a broad range of knowledge and skills students need for approaching the problems frequently encountered when caring for older adults. The book encompasses the continuum of care in various clinical settings, and highlights the geriatric syndromes, common acute and chronic illnesses, as well as other pertinent topics (such as polypharmacy and palliative care). The book also emphasizes the importance of geriatric interdisciplinary team members, including nurses, social workers, advanced practice nurses, therapists, and how these different disciplines intersect in the older adult's care.

The smaller scope of this companion text makes it easier for students to immediately integrate knowledge into clinical application. Cross-referencing will help them find more detailed information in *Geriatric Medicine*, 4th edition, and aid in fostering self-directed learning. Finally, it is our hope that this companion book's case-based instructional approach will help students become familiar with a methodology for dissecting the complexity of disease prevention, presentation, and treatment in older adults. This also gives them a clinical frame of reference in the medical decision-making process and prioritization of diagnostic problems that are so common in the care of older adults.

We are truly indebted to the contributors to *Fundamentals of Geriatric Medicine* and to the authors of the original chapters of *Geriatric Medicine*, 4th edition. We also acknowledge the mentorship and assistance from the associate editors, Rosanne M. Leipzig, MD, PhD, and Christine K. Cassel, MD, MACP, whose original work in *Geriatric Medicine,* 4th edition, made this companion book possible. We are eternally grateful to the John A. Hartford Foundation, Inc. and most especially, to William T. Comfort, Jr., trustee, for their invaluable support and vision for this book. Lastly, we also thank James O'Sullivan, Robert Albano, Sadie Forrester, Barbara Chernow, Suzy Goldhirsch, Sarah Panepinto, Jennifer Reyes, Gerard

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Introduction

Modern educators understand that adults learn in many different ways. Perhaps the least effective is to plough through large, densely packed scholarly textbooks. And, yet, these large textbooks remain the standard of collecting state-of-the-art information for the education of medical students, residents, fellows, and physicians in practice. They are not books to be read through, but books to be used as references. Soriano has created a book that exemplifies a modern approach to adult education. Based on the reference text, it is, importantly, case-based and includes just in time learning and brings together related areas. The case-based nature is what really captures medical learners. Physicians in practice are said to be able to incorporate information much more easily if it is directly related to a patient they are currently seeing. The same, of course, is true for medical students and residents. What this book offers are realistic, clinically based cases that draw the reader into the practical realities of patient management and then provide state-of-the-art, evidence-based information that can contribute to learning about how to care for that kind of patient. In addition, instead of taking each pathophysiologic or organ-system condition on a one-by-one basis as is traditional in scholarly textbooks, Soriano has pulled together related areas that often come together clinically; for example, delirium, dementia, and depression. These three would be separate chapters in a major scholarly textbook, and yet often are overlapping and connected in clinical practice.

In addition to providing clinically relevant information in a way that is engaging and easy to absorb, Soriano has recognized that there are new ways of delivering medical care, especially for the geriatric patient. That is to say, it is not just the physician—certainly not just the geriatrician—whose expertise is relative to the care of the geriatric patient. The geriatrician, other primary care physicians, medical and surgical specialists, nonphysician providers, nurses, nurse practitioners, physical therapists, social workers, pharmacists, and many others have direct contact with patient care. All of those professionals will find this book accessible and relevant to the work they do and to the learning necessary to provide the best care for their patients. In addition to clinical knowledge, Soriano has added a heavy dose of tips on interpersonal and communication skills: one of the most challenging aspects of geriatric medicine. In addition to managing multiple complex chronic illnesses, often with acute illnesses superimposed, and multiple interacting medications and the complications of hospitalizations and procedures, there are also complex family and personal issues around health care decision making. *Fundamentals of Geriatric Medicine* walks the student or the practitioner through these kinds of challenging communication situations in an educational mode that is as effective as their conveying of clinical information.

Why is this book so different from other textbooks? One of the reasons is that Rainier Soriano is a clinician who has extensive experience taking care of patients who present with complex illnesses and need continuity of care. He has taken that experience and applied it to a textbook that is eminently practical in the teaching of medical students, residents, fellows, and physicians in practice.

This may be a new model of textbooks for the future; and if so, it is a good one.

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1
Approach to the Older Adult Patient

Rainier P. Soriano

Learning Objectives

Upon completion of the chapter, the student will be able to:

1. Identify the different components of the history and physical examination and how these differ in older adults compared to younger adults.
2. Identify and understand the potential challenges in caring for older adults and ways to overcome them.
3. Enumerate the changes that occur with normal aging and contrast them with changes that occur secondary to disease.

Case (Part 1)

You are working with Dr. Hopkins, a primary care physician in the community. He has a very busy practice and most of the patients are older adults. You meet him in his office. Upon entering, you notice how brightly lit the entire space is. Large signs hang around the office space. You also noticed that the patient information handouts have a print size that is unusually large, and the print is in black and on white paper. You think it is a little plain compared to the multicolored patient information handouts with fancy fonts on glossy paper you saw at the pediatrics practice you were in last week.

Dr. Hopkins meets you and brings you over to the examining room where a patient is waiting.

Material in this chapter is based on the following chapters in Cassel CK, Leipzig RM, Cohen HJ, Larson EB, Meier DE, eds. Geriatric Medicine: An Evidence-Based Approach, 4th ed. New York: Springer, 2003: Taffet GE, Physiology of Aging, pp. 27–35. Tangarorang GL, Kerins GJ, Besdine RW. Clinical Approach to the Older Patient: An Overview, pp. 149–162. Koretz B, Reuben DB. Instruments to Assess Functional Status, pp. 185–194. Reuben DB. Comprehensive Geriatric Assessment and Systems Approaches to Geriatric Care, pp. 195–204. Selections edited by Rainier P. Soriano.

General Considerations

The initial evaluation of older patients with multiple disorders and treatments is generally prolonged, as compared with the time needed for younger persons. Brief screening questions, rather than elaborate instruments, are appropriate for first encounters (1); more detailed assessment should be reserved for patients with demonstrated deficits (2). Dividing the new patient assessment into two sessions can spare both patient and physician an exhausting and inefficient 2-hour encounter. Other office personnel can collect much information by questionnaire before the visit, from previous records, and from patient and family prior to the physician's contact. It is essential that good care, fully informed by current geriatrics knowledge, be delivered within a reasonable time allocation consistent with contemporary patterns of primary care. One hour for a new visit and 30 minutes for a follow-up are an absolute maximum in most environments.

Completing a home visit may also provide valuable insight into a patient's environment and daily functional status. How mobility may affect function in a particular environment, nutrition, medication use and compliance, and social interactions and support can all be assessed quickly by a home visit. Comprehensive geriatric evaluation and management by an interdisciplinary team in selected populations may improve overall health outcomes, maintain function, and possibly reduce health care utilization (3,4).

Sites of Care

Ambulatory Office Setting

The common occurrence of physical frailty among older persons demands particular attention to providing both a comfortable and safe environment for evaluation. Autonomic dysfunction is commonly encountered in older persons and increases vulnerability to excessively cool or warm settings, especially when the patient is dressed appropriate to the outside temperature. Accordingly, examining rooms should be kept between 70° and 80°F. Brighter lighting is required for adequate perception of the physician's facial expression and gestures by the older patient, whose lenses admit less than half the light they did in youth, due to cross-linking of lens proteins.

Presbycusis (present in >50% of older persons) makes background noise more distracting and interferes with the patient's hearing. Even in a quiet setting, the high-tone loss of presbycusis makes consonants most difficult to discriminate; speaking in a lower-than-usual pitch helps the patient hear, and facing the patient directly improves communication by allowing lip reading. The patient's eyeglasses, dentures, and hearing aid should always be brought to and used at the physician visit. Chairs with a higherthan-standard seat or a mechanical lift to assist in arising are useful for frail older persons with quadriceps weakness, and a broad-based step stool with handrail can make mounting and dismounting the examining table safe. Drapes for the patient should not exceed ankle length so as not to be a risk for tripping and falling.

Acute Hospital or Nursing Home Setting

The patient room is commonly the site of evaluation for the nursing home resident or hospitalized older adult. There is little difference in evaluating older persons in the hospital; the patient is usually confined to bed, so that safety and comfort are dictated by hospital amenities. All other considerations relevant in the ambulatory setting apply. Respect demands either drawing the privacy curtain or, in the nursing home, asking a roommate to leave, if possible. Privacy and identification of the nursing home room as a living space rather than a medical-care space are important issues for the nursing home resident and staff.

Case (Part 2)

Dr. Hopkins introduces you to Mrs. Bauer, a 78-year-old woman with hypertension and osteoarthritis. Dr. Hopkins tells Mrs. Bauer that you will be taking her history while he finishes up with another patient in another room. As soon as Dr. Hopkins leaves the room, Mrs. Bauer turns to you and remarks how she is amazed how young you look. She even comments that you remind her so much of her grandchild. You smile back at Mrs. Bauer and as you take a deep breath, you then proceed to take Mrs. Bauer's history.

The History

Although it is important to discover the patient's reliability as soon as possible, one should not simply dismiss patients with dementia as unreliable and confine data collection to other informants or previous records (1). Beginning history-taking with questions whose answers will illuminate mental status (e.g., time and place orientation, reasons for the visit, previous health care contacts, biographical data, problem solving) can quickly establish the credibility of responses. Even in cases of severe impairment, questions concerning current symptoms may still give useful information (5), and interaction at any level is an essential part of conveying a caring and respectful attitude.

Regardless of mental status, it is common for older patients to be accompanied by family members. Always give the patient the option of being interviewed and examined alone; including family members or companions during the visit should only occur at the patient's request. Certain older adults may be more comfortable meeting the physician with others present, but this decision should be left to the patient.

Generally, it is best to ask if the patient wants a relative or other concerned person present for the history or physical examination; getting some time alone with the patient is essential for the patient to communicate any information he or she regards as confidential for the physician (6). If the relative is present during the history, it is critical to make clear that the patient is to answer all questions; the relative should answer only if the physician asks for clarification. In cases of cognitive impairment or a long and complex history, family members, previous medical records, and other providers can provide supplementary data.

The Chief Complaint

A single or chief complaint is less common among older patients; more often, multiple diseases and problems have multiple symptoms and complaints associated with them. Accordingly, trying to structure the history in the standard format of chief complaint, history of present illness, and past medical history usually results in frustration for the clinician. More useful is the enumeration of a comprehensive problem list, followed by complaints, recent and interval history, and remote information for each of the active problems being considered at a visit. The first evaluation, regardless of setting, requires creating a complete database; its future utility justifies the initial time allocation. It may be easiest to commence the history with an open-ended question, such as, "What do you feel interferes most with your day-to-day activities?" Such a question is usually very helpful in focusing the clinical evaluation.

Another common finding in older patients is that the *law of parsimony*, or *Occam's razor*, is not valid—multiple complaints and abnormal findings arise from multiple diseases; discovering a diagnosis that unifies multiple signs, symptoms, and laboratory data is uncommon and, although welcome in patients of any age, should not be expected in older persons. The classic paradigm in which clinical findings lead directly to a unifying diagnosis has been found to operate in fewer than half of older patients studied (7).

Case (Part 3)

As you proceed with Mrs. Bauer's interview, you start asking her about her medications. She says that she takes enalapril 5 mg daily and acetaminophen 325 mg every 6 hours as needed for her joint pains. She start to bring out her bottles of medications and you notice that she had two of the same medication in two separate bottles. Mrs. Bauer also says that she takes several vitamins and minerals but did not bother to bring them since, according to her, they are "not medications" anyway.

Medication History

The importance of collecting and inquiring about each and every medication taken by or in the possession of the older patient cannot be overemphasized (8). Older adults often take duplicate, overlapping, and conflicting drugs (9), usually acquired from multiple prescribing physicians and over-the-counter sources. All drugs owned by the patient, including supplements, herbals, vitamins, laxatives, sleeping pills, and cold preparations, should be gathered from the bathroom cabinet, bedside table, purse, kitchen drawer, and relatives, and brought to the office visit. Ask patients specifically about food and vitamin supplements and the use of any other alternative medications or remedies they may take. Also, a sizable number of people who are taking alternative medicines fail to inform the clinician, which emphasizes the need for the clinician to specifically inquire about nonprescription drug use (10). All containers should be placed on a table and the patient asked how often each drug is taken and for what, if any, symptomatic indication (8). Consider review of all types of medications every 3 months. Ascertaining pneumococcal, influenza, and tetanus vaccination status can be conveniently done as part of the medication history.

In the hospital, caution should be exercised in ordering all drugs that have been prescribed; toxic accumulation of one or several agents is common, usually because the patient has not been taking prescribed medications as instructed. Self-protective nonadherence to the regimen, often in response to adverse reactions when drugs were taken as ordered, has led to reduction in dosage or frequency by the patient. Return to the originally prescribed schedules produces toxicity.

In the nursing home, the major additional caution concerns the continued administration of unnecessary drugs that were initiated for transient problems arising during hospitalization. Sedative, antipsychotic, diuretic, antiarrhythmic, and antiinfective drugs are often continued indefinitely at the nursing home on the incorrect assumption that they are needed (11). Careful winnowing of the medication list is indicated at least every 3 months and following any hospitalization.

Social History

Crucial information for developing a coherent and feasible care plan at home includes any change in living arrangements, who is available at home or in the local community, and what plans if any exist for coverage in times of illness or functional decline. Although a home visit is the best way to evaluate risks or limitations, inquiring about stairs, rugs, thresholds, bathing facilities, heating, and neighborhood crime can increase the care plan's utility. Stable and durable plans for care at home fulfill both the patient's goal to remain at home and the system's goal to control costs of institutional care. Extent of social relationships is a powerful predictor of functional status and mortality for older adults (12); accordingly, determining the patient's friendship network and recommending ideas for increased socialization can be an appropriate clinical role. Encouraging older persons to become involved with local senior center activities may be one mechanism to enhance social relationships, reduce isolation, and improve daily functioning.

Nutrition History

Although independent elders in the community are generally adequately nourished, the prevalence of under- and overnutrition increases in older persons. Undernutrition is most often unrecognized (13). Most of the undernutrition occurs in those with chronic diseases that directly or indirectly interfere with nutrition. Oral or gastrointestinal disorders, drug effects on appetite, systemic illness, and psychiatric disease increase the risk of undernutrition. Screening questions include a diet history (within the past few days), pattern of weight during recent years, and shopping and food preparation habits (14). Other questions should include any recent intentional efforts to gain or lose weight or any history of eating disorders. Sites of eating, companionship, and skipped meals are also relevant. Serum albumin is a good marker of nutritional status over the preceding 3 months and is correlated with mortality rates (15). The prealbumin reflects nutrition over the past 20 days.

Though not strictly part of the nutritional history, this is a convenient time to inquire about alcohol and tobacco use. Alcohol misuse by older persons is often overlooked in all but florid situations, in part because symptoms and signs are often attributed to other problems common in older persons (16). The CAGE (*c*ut down, *a*nnoyed, *g*uilty feelings, *e*ye-opener) questionnaire has been validated in older persons (17). It is more useful than the MAST-G (Michigan Alcoholism Screening Test–geriatric version) as it requires only four easily memorized questions (18) and has comparable sensitivity and specificity (70% and 81%, respectively, for MAST-G score ≥5 and 63% and 82%, respectively, for CAGE score ≥2). A positive response to two of four questions has traditionally been considered a positive screen. However, in a patient population with high prevalence of drinking problems, even a score of 1 should trigger appropriate investigation (19).

Family History

Although causes of mortality among relatives are usually irrelevant, history of Alzheimer's disease or nonspecified dementia appears to be important. Likewise, certain psychiatric disorders, such as depression and dysthymia, appear to cluster within families. From a mental health perspective, the medical family history also can identify caregiving by the older person.

Caregiving of a disabled spouse and its attendant stress confers substantial mortality risk for the caregiver. Persons who were providing care and who were experiencing strain have a 50% greater risk of dying within 4 years, compared to noncaregiving controls (20).

Sexual History

Older persons continue to be sexually active unless inhibited by the absence of a partner or the occurrence of a disease that reduces libido, makes intercourse painful, or prevents it mechanically (21). Discomfort or awkwardness may result from physician rather than patient attitudes; a simple open-ended question, such as "Tell me about your sex life" or "Are you satisfied with sex?" may encourage the older person to give information not spontaneously reported.

Miscellaneous History

Routine questions should be asked regarding driving habits, seatbelt use, recreational activities, and gambling history.

Case (Part 4)

Mrs. Bauer says that she is able to care for herself. In fact, one of her sons actually moved back in with her. She even gets up early in the morning to cook hot cereal for him. She says that she often does his laundry as well as her own. She cleans up her own apartment but only if her joints are not hurting that much. She regularly swims at a local YMCA but has to take a short bus ride downtown to get there.

Self-Reported Functional Status

The patient should be asked screening questions about independence and self-care—ability to get out of bed, dress, shop, and cook (22,23). Any reported or observed difficulty should provoke more elaborate questions concerning dependence in basic activities of daily living (ADLs), that is, mobility, bathing, transferring, toileting, continence, dressing, hygiene, and feeding (24) and in instrumental activities of daily living (IADLs), that is, shopping, cooking, cleaning, managing money, telephoning, doing laundry, and traveling out of the house (25). Questions should also be asked about vision, hearing, continence, and depression; deficits should be followed up.

Values History (Preferences for Care)

It is wise to take a values history—the patient's beliefs about technological interventions to prolong life, what defines life quality for the patient as an individual, and with what decrements the patient would still think life were worth living. Documenting discussions, executing a living will, and designating a proxy decision maker and durable power of attorney for health care are part of this process of helping the patient have a voice in decisions that may need to be made when the patient, by reason of illness, cannot participate (26).

Physical Examination

General Appearance

General appearance of the older patient should include any noteworthy features; vitality, markedly youthful or aged appearance, and any indicators of frailty or clinical problems (e.g., odor of urine or stool; signs of abuse, neglect, or poverty; hygiene and grooming) warrant mention. Merely observing how long it takes for the patient to get ready for the examination and the extent and nature of help that may be required remains a useful and reliable tool to measure functional capacity (Table 1.1).

Vital Signs

Vital signs do not change with age. Hypothermia is more common, and reliable low-reading thermometers are essential, especially for emergency room and wintertime use. Blood pressure should be taken with the patient in the supine position after at least 10 minutes' rest, and immediately and 3 minutes after standing. Orthostatic hypotension defined as either a 20 mmHg drop in systolic pressure or any drop accompanied by typical symptoms occurs in 11% to 28% of individuals older than 65 years (27,28). In acute moderate blood loss, postural hypotension is a fairly specific but poorly sensitive sign of hypovolemia (29). Blunting of the baroreflex mechanism with age makes cardioacceleration with the upright position a late and unreliable sign of volume depletion in older persons.

Tachypnea at a rate of more than 25 per minute is a reliable sign of lower respiratory infection, even in very elderly patients (30). Weight is the most reliable measure of undernutrition in older outpatients and should be carefully recorded under comparable conditions at each visit. Specific assessment of general or localized pain should be considered as the *fifth vital sign* and should be recorded using a uniform scale (e.g., 0 to 10) (31).

Table 1.1. Major changes in systems

Endocrine system	Impaired glucose tolerance (fasting glucose increased 1 mg/dL/decade; postprandial increased 10 mg/dL/decade)
	Increased serum insulin and increased hemoglobin ($HgbA_{1C}$)
	Nocturnal growth hormone peaks lost; decreased insulin-like growth factor 1 (IGF-1)
	Marked decrease in dehydroepiandrosterone (DHEA)
	Decreased free and bioavailable testosterone
	Decreased triiodothyronine (T_3)
	Increase parathyroid hormone (PTH)
	Decreased production of vitamin D by skin
	Ovarian failure, decreased ovarian hormones
	Increased serum homocysteine levels
Cardiovascular	Unchanged resting heart rate (HR); decreased maximum HR
	Impaired left ventricular filling
	Marked dropout of pacemaker cells in sinoatrial (SA) node
	Increased contribution of atrial systole to ventricular filling
	Left atrial hypertrophy
	Prolonged contraction and relaxation of left ventricle
	Decreased inotropic, chronotropic, lusitropic response to β-adrenergic stimulation
	Decreased maximum cardiac output
	Decreased hypertrophy in response to volume or pressure overload
	Increased serum atrial natriuretic peptide
	Large arteries increase in wall thickness, lumen, and length, and become less distensible and compliance decreases
	Subendothelial layer thickened with connective tissue
	Irregularities in size and shape of endothelial cells
	Fragmentation of elastin in media of arterial wall
	Peripheral vascular resistance increases
Blood pressure (BP)	Increased systolic BP, unchanged diastolic BP
	β-Adrenergic–mediated vasodilatation decreased; α-adrenergic–mediated vasoconstriction unchanged
	Brain autoregulation of perfusion impaired
Pulmonary	Decreased forced expiratory volume at 1 second (FEV_1) and forced vital capacity (FVC)
	Increased residual volume
	Cough less effective
	Ciliary action less effective
	Ventilation-perfusion mismatching causes PaO_2 to decrease with age: $100 - (0.32 \times \text{age})$
	Trachea and central airways increase in diameter
	Enlarged alveolar ducts due to lost elastic lung parenchyma structural support result in decreased surface area
	Decreased lung mass
	Expansion of thorax
	Maximum inspiratory and expiratory pressures decrease
	Decreased respiratory muscle strength
	Chest wall stiffens
	Diffusion of CO decreased
	Decreased ventilatory response to hypercapnia
Hematologic	Bone marrow reserves decreased response to high demand
	Attenuated reticulocytosis to erythropoietin administration
Renal	Decreased creatinine clearance and glomerular filtration rate (GFR) 10 mL/decade

TABLE 1.1. *Continued*

	25% decrease in renal mass mostly from cortex with a relative increased perfusion of juxtamedullary nephrons
	Decreased sodium excretion and conservation
	Decreased potassium excretion and conservation
	Decreased concentrating and diluting capacity
	Impaired secretion of acid load
	Decreased serum renin and aldosterone
	Accentuated antidiuretic hormone (ADH) release in response to dehydration
	Decreased nitric oxide production
	Increased dependence of renal prostaglandins to maintain perfusion
	Decreased vitamin D activation
Genitourinary	Prolonged refractory period for erections for men
	Reduced intensity of orgasm for men and women
	Incomplete bladder emptying and increased postvoid residuals
	Decreased prostatic secretions in urine
	Decreased concentrations of antiadherence factor Tamm-Horsfall protein
Temperature regulation	Impaired shivering
	Decreased cutaneous vasoconstriction and vasodilation
	Decreased sweat production
	Increased core temperature to start sweating
Muscle	Marked decrease in muscle mass (sarcopenia) due to loss of muscle fibers
	Aging effects smallest in diaphragm (role of activity), more in legs than arms
	Decreased myosin heavy chain synthesis
	Small if any decrease in specific force
	Decreased innervation, increased number of myofibrils per motor unit
	Infiltration of fat into muscle bundles
	Increased fatigability
	Decrease in basal metabolic rate (decrease 4%/decade after age 50) parallels loss of muscle
Bone	Slower healing of fractures
	Decreasing bone mass in men and women both trabecular and cortical bone
	Decreased osteoclast bone formation
Joints	Disordered cartilage matrix
	Modified proteoglycans and glycosaminoglycans
Peripheral nervous system	Loss of spinal motor neurons
	Decreased vibratory sensation especially in feet
	Decreased thermal sensitivity (warm and cool)
	Decreased sensory nerve action potential amplitude
	Decreased size of large myelinated fibers
	Increased heterogeneity of axon myelin sheaths
Central nervous system	Small decrease in brain mass
	Decreased brain blood flow and impaired autoregulation of perfusion
	Nonrandom loss of neurons to modest extent
	Proliferation of astrocytes
	Decreased density of dendritic connections
	Increased numbers of scattered neurofibrillary tangles
	Increased numbers of scattered senile plaques
	Decreased myelin and total brain lipid
	Altered neurotransmitters including dopamine and serotonin
	Increased monoamine oxidase activity
	Decrease in hippocampal glucocorticoid receptors

TABLE 1.1. *Continued*

	Decline in fluid intelligence
	Slowed central processing and reaction time
Gastrointestinal	Decreased liver size and blood flow
	Impaired clearance by liver of drugs that require extensive phase I metabolism
	Reduced inducibility of liver mixed-function oxidase enzymes
	Mild decrease in bilirubin
	Hepatocytes accumulate secondary lysosomes, residual bodies, and lipofuscin
	Mild decrease in stomach acid production, probably due to nonautoimmune loss of parietal cells
	Impaired response to gastric mucosal injury
	Decreased pancreatic mass and enzymatic reserves
	Decrease in effective colonic contractions
	Decreased calcium absorption
	Decrease in gut-associated lymphoid tissue
Vision	Impaired dark adaptation
	Yellowing of lens
	Inability to focus on near items (presbyopia)
	Minimal decrease in static acuity, profound decrease in dynamic acuity (moving target)
	Decreased contrast sensitivity
	Decreased lacrimation
Smell	Detection decreased by 50%
Thirst	Decreased thirst drive
	Impaired control of thirst by endorphins
Balance	Increased threshold vestibular responses
	Reduced number of organ of Corti hair cells
Audition	Bilateral loss of high frequency tones
	Central processing deficit
	Difficulty discriminating source of sound
	Impaired discrimination of target from noise
Adipose	Increased aromatase activity
	Increased tendency to lipolysis
Immune system	Decreased cell-mediated immunity
	Lower affinity antibody production
	Increased autoantibodies
	Facilitated production of antiidiotype antibodies
	Increased occurrence of monoclonal gammopathy of unknown significance (MGUS)
	More nonresponders to vaccines
	Decreased delayed type hypersensitivity
	Impaired macrophage function [interferon-γ, transforming growth factor-β (TGF-β), tumor necrosis factor (TNF), interleukin-6 (IL-6), IL-1 release increased with age]
	Decreased cell proliferative response to mitogens
	Atrophy of thymus and loss of thymic hormones
	Accumulation of memory T cells (CD-45+)
	Increased circulating IL-6
	Decreased IL-2 release and IL-2 responsiveness
	Decreased production of B cells by bone marrow

Source: Taffet GE. Physiology of aging. In: Cassel CK, Leipzig RM, Cohen HJ, et al., eds. Geriatric Medicine, 4th ed. New York: Springer, 2003.

Integument

Skin undergoes many changes with age, including dehydration, thinning, and loss of elastic tissue. Wrinkling is more powerfully predicted by sun exposure and cigarette smoking than by age. Most proliferative lesions, benign and malignant, are related to sun exposure; accordingly, basal and squamous cell cancers and melanomas should be most aggressively hunted on exposed skin. Because of skin aging, turgor is not a reliable sign of hydration status. All skin should be examined, exposed to sun or not, for evidence of established or incipient (nonblanching redness) pressure sores. Ecchymoses should also be noted, whether due to purpura of thin old skin or trauma; the possibility of abuse should be considered.

Head and Neck

Head and neck examination begins with careful observation of sun-exposed areas for premalignant and malignant lesions. Palpation of temporal arteries for pain, nodularity, and pulse is recommended, but asymptomatic temporal or giant cell arteritis or polymyalgia rheumatica is not common (32), and palpation is an insensitive test. Arcus ocularis, or cornealis, a white-to-yellow deposit at the outer edge of the iris, along with xanthelasma, predicts premature coronary disease in young adults. Beyond age 60, it does not identify increased risk.

Visual acuity and hearing screening are necessary, given the high prevalence of impaired vision and auditory acuity among older persons. For most clinical situations, a pocket Snellen chart, held 14 inches from the eye, is more practical than a wall-mounted chart. The whispered voice is as sensitive as an audioscope for detection of hearing loss (33,34), but the latter is, to date, the best objective measurement of hearing and more accurate at following changes over time. Inspecting the ear canals and drums using an otoscope is especially necessary if hearing loss is detected; removing impacted cerumen is a common quick-fix intervention for many older patients.

Oral examination for denture sores, tooth and gum health, and oral cancers is essential and should include inspection and palpation with dentures out (35). The earliest detectable malignant oral lesion is red and painless; if persistent beyond 2 weeks, biopsy is mandatory. Although on the decline, oral cancers are most common in older persons with long-standing alcohol or tobacco use or poor hygiene.

Vascular sounds in the neck usually arise from vessels other than the carotid artery (36); true carotid bruits confer more risk for coronary events and contralateral stroke than for ipsilateral stroke, and may cease unpredictably (37).

Breast Examination

Breast examination is generally simpler in older women. Fat diminishes, making breast tissue and the tumors that arise from it more easily palpable. Routine screening mammograms annually or every other year should be continued lifelong or until a decision is reached that a discovered cancer would not be treated (38); age-specific breast cancer incidence increases at least until age 85, and no evidence indicates that treatment is not effective in older women (39,40). Current recommendations for breast cancer screening suggest yearly mammography until age 69, but there has been much discussion about revising the age to 74 or 79, or removing an upper age limit entirely. Routine screening mammography annually is part of Medicare benefit, and age cutoffs or stopping screening on the basis of age alone is controversial. Accordingly, discussions as to how the information will be utilized should take place before testing is initiated.

Case (Part 5)

After your thorough history taking, you asked Mrs. Bauer to change to a hospital gown in preparation for her physical examination. Her blood pressure (BP) = 150/75 mmHg; heart rate (HR) = 62/min; respiration rate (RR) = 12/min; and temperature = 36.7 C. Her head and neck exam is unremarkable. Her lungs are clear to auscultation and percussion. There is a systolic murmur present at the aortic area on exam without any radiation to the neck. Her abdomen and extremities examination are normal.

Heart and Lung Examination

Lung examination is little different in the older patient. Rales are abnormal at any age; evanescent crackles of atelectasis are the most common cause of rales in the absence of pathology.

Cardiac examination has several special features in aged patients. Both atrial and ventricular ectopy are common at baseline without symptoms or ominous prognosis (41,42). Although S_4 is common among older persons free of cardiac disease, S_3 is associated with congestive heart failure. The ubiquitous systolic ejection murmur is less reliable as a sign of hemodynamically significant aortic stenosis in older individuals. A loud murmur (>2/6), diminution of the aortic component of S_2, narrowed pulse pressure, and dampening of the carotid upstroke suggest aortic stenosis, but each may be absent and be falsely reassuring (43). In the absence of typical symptoms, a systolic ejection murmur lacking any of the features of stenosis may be followed without cardiac imaging. Although for decades aortic sclerosis was considered benign, it has recently been associated with increased risk for myocardial infarction, congestive heart failure, stroke, and death from cardiovascular causes, even without evidence of significant outflow tract obstruction (44).

Abdominal and Rectal Examination

Abdominal and rectal examination have few additional or special components for the older patient. Unsuspected fecal impaction is common and, despite no complaint of constipation, should be treated with a bowel regimen that includes fiber and scheduled toileting. Evidence of fecal or urinary incontinence is usually obvious to the alert examiner. A chronically overfilled and distended bladder should be suspected in men who are incontinent. Although part of the screen for prostate cancer, prostatic masses detected on digital rectal examination may also reflect granuloma, calcification, or hyperplasia, and benign causes outnumber malignant ones; differentiation by imaging is thought not to be reliable unless calcification is present. Prostatic enlargement of benign hyperplasia (since cell proliferation occurs, hypertrophy is an incorrect term) correlates poorly with both urethral obstruction and symptoms of prostatism; anterior periurethral encroachment causes symptoms, but it is the posterolateral portions of the gland that are accessible on digital examination. The need for and utility of screening for fecal occult blood in the early detection and reduction of mortality and morbidity of colon cancer is established in patients of all ages (45,46).

Musculoskeletal Examination

Examination of the musculoskeletal system, which is often a source of abundant complaints and pathology in older adults, begins with simple screening. In the absence of complaints or loss of function, brief tests of function are adequate to reveal unsuspected limitations. For the upper extremity, asking the patient to "touch the back of your head with your hands" and "pick up the spoon" are sensitive and specific (1). Gait and mobility can be assessed by the timed "up-and-go" test (arise from a chair, walk 3 meters, turn, walk back, and sit down) (47); requiring that each foot be off the floor in the up-and-go test makes the test a better predictor of functional deficits than a standard detailed neuromuscular examination (48). Neuromuscular abnormalities may not identify persons with mobility deficits and demonstrable difficulties in the up-and-go test. When deficits are detected in any screening test, more detailed evaluation, including neuromuscular exam and longer standard objective tests (2,49), and likely inclusion of a physical therapist in evaluation and treatment are indicated.

Pelvic Examination

Pelvic examination in older women often is neglected. Atrophic vaginitis, with associated urinary incontinence, or itching or dyspareunia, is remarkably easy and gratifying to treat. Ovaries or uterus palpable more than 10 years beyond menopause usually indicate pathology, often tumor. Any adnexal mass in a woman over 50 years is considered malignant until proven otherwise (50). If arthritis or frailty make stirrups uncomfortable for the patient, examination in bed or on a table with the patient positioned on her side with knees drawn up will allow speculum exam and Papanicolaou smear. The bimanual exam can be done with the patient supine, again avoiding use of stirrups. Signs of abuse may only be apparent on pelvic examination.

Neurologic Examination

The most important principle is that although abnormalities are common in the neurologic examination of the older patient, one-third to one-half the abnormal findings have no identifiable disease causing them. Abnormalities are classified as (1) attributable to a disease or an isolated abnormality, and (2) more common with increasing age or not. Abnormalities occurring in the absence of detectable disease and more common with increasing age are the best current definition of changes of aging in the nervous system. Abnormalities attributable to disease and more common with increasing age simply reflect diseases that are more common in older persons and have nervous system findings. Abnormalities occurring in the absence of detectable disease but not more common with increasing age are most likely individual variations not attributable to aging; the unlikely possibility also exists that lack of progression occurs following changes that developed before age 65.

The considerable prevalence of neurologic abnormalities in older persons carefully evaluated and found to have no disease explaining the finding demands even greater caution in attributing predictive significance to the abnormality. For example, *frontal release signs* (also called primitive reflexes)—snout, palmomental, root, suck, grasp, glabellar tap—have been reported to identify patients with dementia (51–53) or with Parkinson's disease. Because these signs appear in 10% to 35% of older adults screened to exclude disease (54–56), it is difficult to accept reports of these signs as identifiers of disease, at least in older persons. Ankle jerks, reported to be absent among many otherwise healthy older persons, turn out to be just a bit more difficult to elicit. Using a high-quality, round neurologic hammer rather than a lightweight, red triangulated hammer, and striking briskly will improve accuracy. It appears that reports of loss of ankle jerk with age may be a result of the care and expertise with which the reflex is elicited, rather than an aging effect (54,57).